Operator-Entropy Joint-Embedding Predictive Architectures for Multivariate Time-Series Anomaly Detection: Latent Residual Conditioning and Calibrated Thresholding

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Abstract*—We propose the Operator-Entropy Joint-Embedding Predictive Architecture (OpEntropy JEPA) for multivariate time-series anomaly detection in high-dimensional sensor telemetry. Deep reconstructive detectors minimize observation-space error, accumulating stochastic sensor noise that distorts residual distributions and causes severe false-alarm flooding under automated threshold calibration. OpEntropy JEPA addresses this limitation by forecasting future telemetry states in an abstract representation space using a linear Koopman transition operator regularized by scale-free Von Neumann spectral entropy and a logarithmic Bregman energy barrier against subspace collapse. Under chronological held-out extreme-value calibration across 33 telemetry streams from seven physical datasets, OpEntropy JEPA suppresses false alarms by four- to five-fold compared with reconstructive baselines, lowering nominal false-positive rates from 30.0% (TimesNet) and 30.7% (TranAD) to 7.6%–8.6% (paired reductions exceeding 0.14, $p < 0.01$). To support continuous streaming, the framework integrates causal coordinate saliency gating and multiscale dyadic frequency shells. We evaluate threshold transfer under retrospective buffered and zero-lookahead causal streaming. An exchangeable permutation protocol establishes that inter-block distribution drift accounts for 84% to 96% of static calibration risk overshoot, dominating sample scarcity and serial autocorrelation. Across 19 multi-seed streams, point-level F1 and precision-recall ranking remain statistically comparable to reconstructive baselines, and physical regularizers confirm Two One-Sided Tests (TOST) equivalence to unconstrained latent prediction within a 0.05 margin ($p < 0.006$). OpEntropy JEPA provides a well-conditioned representation space that resolves false-alarm flooding under automated operational calibration.

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In complex deployments (NASA SMAP, MSL rovers, industrial IoT, and biometric arrays), dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], where faults must be identified before hardware failure occurs.

Catastrophic False Alarms in Reconstructive Baselines: Multi-sensor monitoring requires automated decision rules that detect physical faults while constraining false alarms. Prevailing deep detectors (TimesNet, TranAD, Autoencoders) minimize point-level reconstruction errors in raw observation space [4], [5]. Because raw signals contain ambient jitter, reconstructive objectives allocate capacity to fitting uninformative high-frequency sensor noise. Under automated extreme-value thresholding such as Peaks-Over-Threshold (SPOT) [6], the resulting heavy-tailed residuals trigger false alarms across 20%–30% of nominal timesteps, flooding operators and rendering alerting unviable.

The Operator-Entropy JEPA Architecture: To eliminate observation-space noise accumulation without artificial contrastive perturbations [7], we propose the *Operator-Entropy Joint-Embedding Predictive Architecture* (OpEntropy JEPA). Rather than reconstructing raw waveforms, the architecture forecasts future target representations in abstract latent space [8]. To maintain modal expressivity and prevent dimensional collapse, OpEntropy JEPA models latent transitions via a linear Koopman operator regularized by scale-free Von Neumann spectral entropy and a logarithmic Bregman energy barrier. We also evaluate hydrodynamic (Reynolds-stress) and field-theoretic (potential-flow) regularizers as comparative controls. For streaming execution, the framework integrates causal coordinate saliency gating, multiscale dyadic shells, and extreme-value tail calibration.

A. Contributions

This work makes four principal contributions:

1) **Operator-Entropy JEPA Formulation:** We formulate OpEntropy JEPA for multi-sensor telemetry, combining latent representation forecasting with a Koopman transition operator regularized by scale-free Von Neumann spectral entropy and an anti-collapse Bregman energy barrier. We establish theoretical guarantees against operator collapse (Proposition 1) and characterize its latent coordinate geometry.

2) **False-Alarm Suppression Against SOTA Baselines:** Under chronological held-out calibration on 19 non-GHL streams, OpEntropy JEPA suppresses false alarms four-fold against reconstructive baselines, achieving 8.6% nominal held-out FPR versus 30.0% for TimesNet and 30.7% for TranAD (paired reductions of -0.144 and -0.142, $p = 0.006$). Bounded linear and convolutional controls confirm that latent score geometry compresses residual tails while maintaining competitive Point-F1 and PR-AUC.
3) **Threshold Calibration Innovations and Drift Decomposition:** We formalize operational thresholding under chronological streaming, showing that static extreme-value calibration systematically overshoots design risk (5.3–6.2% vs. $\approx 22\%$ buffered; 1.0% vs. 16–18% causal at $q{=}10^{-3}$). Under an exchangeable permutation protocol, inter-block distribution drift accounts for 84–96% of risk overshoot, dominating sample scarcity and autocorrelation.
4) **Reproducible Multi-Seed Benchmark and Null Results:** Across 33 continuous streams from seven physical datasets (19 non-GHL at five seeds; 14 GHL at one to three seeds), we evaluate causal and buffered streaming regimes. Across fixed-backbone controls, physical regularizers confirm formal Two One-Sided Tests (TOST) equivalence to unconstrained latent prediction within ± 0.05 Point-F1 ($p \leq 0.0054$). All code, data, and scripts are released open-source.

II. Related Work

Dynamical Systems, Reconstruction, and Forecasting Detectors: Multi-sensor telemetry models system dynamics as state trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [9]. Neural ODEs [10] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness. Koopman theory and dynamic mode decomposition [11]–[13] learn linear operators on observable embeddings, identifying disruptions via spectral decoherence; our proposed Operator-Entropy JEPA architecture transfers this principle to a latent transition operator regularized by spectral entropy.

Reconstructive detectors model nominal patterns via observation reconstruction: LSTM forecasting with dynamic thresholding for spacecraft telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity variation in TimesNet [5], and adversarial sub-series attention in TranAD [14]. These share a vulnerability: point-level losses bias models toward stochastic noise, accumulating residual variance across irrelevant channels in high-dimensional arrays.
Threshold Calibration, Contrastive, and Self-Supervised Detection: Unsupervised monitoring requires automated rules to map residual scores to binary alarms without labels. Non-parametric thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [6] fit Generalized Pareto tails under design risk q; however, EVT asymptotic guarantees assume i.i.d. samples, which temporal autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods train discriminative representations: NCAD [7] injects synthetic anomalies, yielding lower calibrated detection than predictive models (0.058 vs. 0.160 macro Point-F1, Supplementary Table S28). DCdetector [15] maximizes dual-attention consistency. Joint-Embedding Predictive Architectures [8], [16], [17] predict target representations in latent space with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] evaluate ranking or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration.
Physics-Inspired Priors and Conceptual Positioning: PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed dynamics as rigid constraints; their stiff competing gradients [25] and PDE requirements fit poorly to uncalibrated telemetry. We distinguish *physics-inspired* structures (exploratory regularizers; latent coordinates are not physical state variables) from *physics-informed* constraints; *causal preprocessing* from *causal decision-time access*; and operator non-degeneracy from representation collapse. This work formulates OpEntropy JEPA and evaluates latent prediction alongside physical controls under held-out nominal calibration to quantify the gap between design risk and realized false alarms.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP Earth-observation `A-1`/`A-2`/`A-7`/`D-3`/`P-3`/`E-1` power, thermal, and attitude subsystems; MSL rover `C-1`/`C-2`/`D-14`/`D-15`/`D-16`/`M-1` actuator and environmental channels, with two short-nominal streams reserved; SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so $S{=}64$ is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample calibration, each stream's nominal sequence is partitioned chronologically into training (65%), calibration (17.5%), and held-out evaluation (17.5%) blocks separated by an isolation gap (Section V-A). The held-out protocol spans 33 streams across seven datasets (6 Daphnet, 14 GHL, 4 MSL, 6 SMAP, Genesis, Metro, SWAN); the remaining 12 streams are reserved as operational case studies (CalIt2, two Room-Occupancy, two short-nominal MSL, GECCO, and six CICIDS) because their nominal blocks

are too short for disjoint partitions ($N_{\text{cal}} < 500$) or, for the 80-channel CICIDS network flows, high-dimensional stride-1 evaluation across multi-seed runs was computationally prohibitive compared to lower-dimensional physical telemetry. The dyadic front-end ablation (Table Ib) uses a separate five-stream panel that additionally draws one SMD channel outside this manifest.

Two caveats bound interpretation. SMAP and MSL contain documented mislabeled segments and near-trivial anomalies [32]; their per-dataset numbers are upper bounds on detectability. Entity coverage is also incomplete (6 of 55 SMAP and 4 of 27 MSL channels, selected by nominal-block length and stride-1 runtime constraints, not labels), so claims apply to the evaluated entities, not the full archives.

Supplementary Table S15 and the benchmark stream manifest enumerate the suite: 2,079,507 nominal training and 2,986,574 test timesteps (>5.06M observations across 33 primary and 12 case-study streams), recording channel dimensions, contamination ratios, anomaly typologies, split lengths, and evaluated seeds. All counts in this paper reconcile against this manifest.

IV. Methodology: Operator-Entropy Joint-Embedding Predictive Architecture

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, which can weight channels with different noise scales unequally; whether this consumes model capacity is not tested here. We instead adopt the JEPA framework [8]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space to filter out stochastic sensor noise and condition prediction residuals for automated tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

We formulate Operator-Entropy JEPA as our primary architecture, and evaluate it alongside an unconstrained baseline (`TS-JEPA`) and two comparative physical formulations (Reynolds-Stress and Potential-Flow):

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [33], [34], where nonlinear trajectories are mapped via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ eigenvalues of $\mathbf{K}\mathbf{K}^\top$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ total energy. For $\mathcal{E}(\mathbf{K}) > 0$, the scale-free modal distribution is $p_i = \sigma_i^2/\|\mathbf{K}\|_F^2$. The Von Neumann spectral entropy is $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i:p_i>0} p_i \log p_i$ (with $0\log 0 = 0$). When all singular values vanish ($\mathbf{K} = \mathbf{0}$), the normalized spectrum is undefined; by boundary convention we define $\mathcal{H}_{\text{spectral}}(\mathbf{0}) \equiv 0$, so that $\Delta\mathcal{H}(\mathbf{0}) = \log D$ attains its maximum deficiency. The *Spectral Entropy Deficiency* measures Kullback-Leibler divergence from the uniform equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0, \quad (1)$$

where $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while dimensional collapse to rank 1 yields $\Delta\mathcal{H} = \log D$. Minimizing $\Delta\mathcal{H}$ drives $\mathbf{K}$ toward an isometric modal equipartition rather than low-rank dominant modal truncation as in classical Koopman decompositions, acting directly on learned embeddings without infinite-dimensional lifting.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) without explosive gain, we impose a non-negative Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r_{\text{safe}}(\mathbf{K}) - 1 - \log r_{\text{safe}}(\mathbf{K}) \geq 0, \quad (2)$$

where $r_{\text{safe}}(\mathbf{K}) = \max(\|\mathbf{K}\|_F^2/(\gamma D), \epsilon)$ with numerical floor $\epsilon = 10^{-6}$. The floored barrier caps penalty at $\approx 12.8\,\beta_{\text{norm}}$ rather than diverging. With $\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$, $r_{\text{safe}} \approx 1$ and training begins with vanishing barrier penalty. The composite regularizer is:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \quad (3)$$

and the training objective minimizes $\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K}z_{\text{ctx}} - z_{\text{tgt}}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K})$, with $\beta_{\text{norm}} = 1.0$ and $\lambda_{\text{ent}} = 0.5$. All shared representation components are held fixed across models and controls. $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ acts on the operator matrix; it does not analytically prevent encoder representation collapse, which is mitigated empirically by EMA target dynamics and stop-gradient operations [16], [35].

Proposition 1 (Operator Non-Triviality and Barrier Behavior)**.** *Let* $\mathcal{L}(\mathbf{K}) = \frac{1}{B}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent} > 0, \beta_{norm} > 0$*, non-trivial target batch* $\frac{1}{B}\sum_{b=1}^{B}\|z_{tgt}^{(b)}\|_2^2 = E_{tgt} > 0$*, and non-degenerate context–target coupling* $\frac{1}{B}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top} \neq \mathbf{0}$*. Then: (i) under the floored barrier* ($\epsilon = 10^{-6}$)*, the energy penalty is bounded* ($\leq 12.8\,\beta_{norm}$) *with boundary convention* $\Delta\mathcal{H}(\mathbf{0}) = \log D$*, yet* $\mathbf{K} = \mathbf{0}$ *is strictly not a local or global minimizer; and (ii) under unfloored idealization* ($\epsilon \to 0^+$)*,* $\lim_{\|\mathbf{K}\|_F\to 0}\mathcal{L}(\mathbf{K}) = +\infty$.

(Proof in Supplementary Section S-1.)

Remark (Operator Non-Degeneracy vs. Representation Collapse): Proposition 1 applies to the transition operator $\mathbf{K}$ and does not analytically prevent encoder collapse: if E_θ maps all inputs to a constant, $\mathbf{K} = \mathbf{I}$ attains zero loss. Preventing representation collapse rests empirically on stop-gradient momentum dynamics.

2) Auxiliary Quadratic Second-Moment Head: Inspired by Reynolds decomposition in fluid turbulence, which separates mean flow from velocity fluctuations, we evaluate an auxiliary quadratic second-moment head in latent space. The observed matrix $\mathbf{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s\times d_s}$ ($d_s = 16$, via projection $\mathbf{P} \in \mathbb{R}^{d_s\times D}$) is an instantaneous

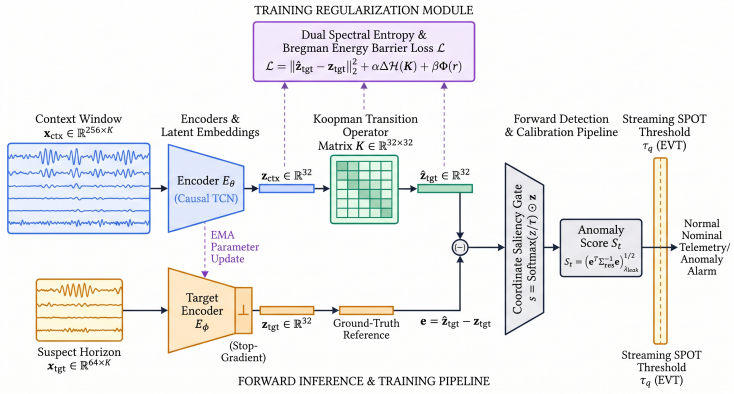


Fig. 1. Operator-Entropy JEPA architecture: multiscale context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and suspect ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) windows are mapped into latent space via online (E_θ) and EMA momentum (E_ϕ) encoders with a stop-gradient barrier ($\perp$). A linear Koopman transition operator ($\mathbf{K} \in \mathbb{R}^{32\times32}$) forecasts the target representation $\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K}\mathbf{z}_{\text{ctx}}$, regularized by dual Von Neumann spectral entropy and a logarithmic Bregman energy barrier against collapse. Residual discrepancies pass through a coordinate saliency gate to yield anomaly score S_t calibrated via streaming SPOT.

rank-1 projected outer product plus ridge penalty, not a physical turbulent covariance. A closure head outputs lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s\times d_s}$ reconstructing predicted covariance $\mathbf{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + 10^{-3}\mathbf{I}$. The objective minimizes $\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1\ \text{MSE}(\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{obs}})$. At test time, scoring adds the second-moment discrepancy: $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}} - z_{\text{tgt}}) + 0.5\, g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$.

3) Potential-Flow JEPA (Conservative Field Theory and Flow Matching): Under Helmholtz-Hodge decomposition, a smooth velocity field splits into a curl-free gradient and a solenoidal component: $v = -\nabla\Phi + \nabla \times A$. We parameterize $\hat{v} = -\nabla_{z_t}\Phi_\psi$ via a residual MLP conditioned on state z_t, diffusion time $t \in [0,1]$, context z_{ctx}, and regime projection p_{regime} selected across $K_r = 4$ orthonormal subspace projections $\mathbf{U}_k \in \mathbb{R}^{D\times d}$ (d=8, D=32), with tangent projection $v_\perp = v - \langle u, v\rangle u$ ($u = z_t/\|z_t\|_2$). Training uses conditional flow matching (OT-CFM) along linear paths between noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and z_{tgt}: $\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}}[\|\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2]$. At inference, the error queries the field at the trajectory midpoint: $\mathbf{e}(t) = \hat{v}(0.5 z_{\text{tgt}},\ 0.5,\ z_{\text{ctx}},\ p_{\text{regime}}) - z_{\text{tgt}}$. Parameterizing $\hat{v}$ as a spatial gradient imposes a curl-free bias ($\nabla \times \hat{v} \equiv \mathbf{0}$) with vanishing circulation $\oint_{\mathcal{C}} \hat{v} \cdot d\mathbf{r} = 0$, hypothesized to penalize cyclic nominal orbits and degrade detection on periodic kinematics (evaluated empirically in Section VII-B). An idealized autonomous continuous gradient flow $\dot{z} = -\nabla\Phi(z)$ admits no non-constant periodic orbits (Proposition S2 in Supplementary Section S-1), providing continuous motivation without binding discrete midpoint scoring.

C. Anomaly Scoring and Operational Detection Latency

During inference, context windows $x_{\text{ctx}} = \mathbf{X}[t-C : t, :]$ and suspect windows $x_{\text{tgt}} = \mathbf{X}[t : t+S, :]$ are processed at stride 1. For TS-JEPA, Operator-Entropy, and Potential-Flow, the score gates the latent discrepancy through $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t))$. For the second-moment head (Reynolds-Stress), $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t)) + 0.5\, g(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}}))$. All shared representation stages are held fixed across variants and ablation controls (Section VI-A).

Three-Part Operational Causality Framework: To resolve the methodological confounding between data availability, decision latency, and evaluation attribution, we formalize detection under three operational criteria (Fig. 2):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision time $\tau = t + S - 1$, the model observes only history $\mathbf{X}_{:\tau} = \mathbf{X}[t - C : \tau, :]$. Neither encoder accesses future observations beyond τ.

(b) **Score-emission latency ($S-1 = 63$ steps):** Because suspect representation z_{tgt} encodes the horizon $\mathbf{X}[t : t+S, :]$, window score s_t cannot be computed until timestep $\tau = t+S-1$. This imposes an inherent emission latency of $S-1 = 63$ steps: 0.98 s on Daphnet (64 Hz), 10.5 min on SMAP, MSL, and GHL under standard 10-s (0.1 Hz) sampling [3], [29], 63 min on GECCO (1 sample/min), and 63 h on Metro (1 sample/h). This latency guarantees zero future lookahead, but must not be confused with instantaneous detection at fault inception: batch scoring over horizon S=64 fits supervisory health monitoring and slow subsystem drift where anomaly persistence distinguishes faults from noise, but cannot provide sub-second emergency tripwires on low-frequency telemetry.

(c) **Alarm attribution convention:** We distinguish two attribution regimes: (1) *strictly causal trailing attribution*, where score s_t is emitted and evaluated strictly at τ, providing zero-lookahead online monitoring; and (2) *retrospective buffered smear attribution*, where s_t is attributed across the entire suspect window $[t+1, t+S]$. Under unadjusted point evaluation, buffered alarms in $[t^* - S, t^*]$ register as false alarms: buffering provides fixed-lag detection, not early warning. The two regimes represent distinct operational realities and are reported separately throughout (Table II, Supplementary Table S31).

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, the latent JEPA architectures incorporate shared inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized

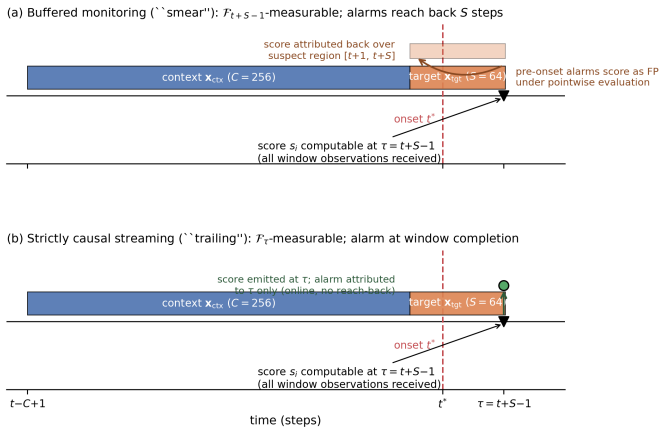


Fig. 2. Score-emission and alarm-attribution timeline. Every score s_i is computable only at window completion $\tau = t+S-1$. (a) Buffered attribution assigns s_i retrospectively to $[t+1, t+S]$; (b) causal attribution emits s_i at τ only.

anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate $g(\mathbf{e}) \equiv \mathcal{D}_{\text{sal}}(\mathbf{e})$ on error vectors $\mathbf{e} = \hat{z}_{\text{tgt}} - z_{\text{tgt}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \qquad (4)$$

$$w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D}\exp(|e_j|/\tau)},$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$. When coordinates share equal absolute deviations ($|e_1| = \cdots = |e_D|$), weights are uniform ($w_d = 1/D$) and $\mathcal{D}_{\text{sal}}(\mathbf{e})$ collapses identically to the Euclidean norm $\|\mathbf{e}\|_2$; under an isotropic Gaussian error distribution $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, coordinate magnitudes fluctuate symmetrically, yielding $\mathbb{E}[\mathcal{D}_{\text{sal}}(\mathbf{e})] \approx \|\mathbf{e}\|_2$ without directional coordinate preference. We fix blend parameter $\alpha = 0.5$ and softmax temperature $\tau = 0.5$ across all latent predictors. Reconstructive baselines (TimesNet, TranAD) evaluate unweighted observation errors and do not use the gate. This is the discrepancy gate used in every latent experiment: input channels are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): We also evaluate a strictly causal *Causal Channel Saliency Gate* as an auxiliary variant (full specification, boundary-condition verification, and the fault-type study in Supplementary Section S-3): each channel is weighted by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to context and target encoders so that $\partial g_k/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (zero target leakage, with an explicit identity short-circuit at $K{=}1$). Two boundary conditions apply: quiet faults (flatlines, stuck values) cannot be isolated by variance amplification (the fault-type study confirms they are weakly detected under *every* gating regime, recall 0.02–0.07), and abrupt regime transitions transiently perturb variance shares until the context window refills.

b) Zero-Leakage Data Provenance and Temporal Isolation: We enforce strict temporal isolation across the training, nominal-calibration, and held-out evaluation partitions (Section V-A): standardization statistics come only from the training partition, EVT tail parameters (u, σ, ξ) only from the calibration partition, and evaluation proceeds chronologically with frozen weights and thresholds. The auxiliary gate's weights depend only on x_{ctx}; the deployed pipeline weights inputs uniformly.

2) Multiscale Frequency Front-End: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore splits each channel into $J{=}4$ smooth dyadic shells on an $N_{\text{fft}} = 256$ rFFT grid, normalized bin-wise into an exact amplitude partition of unity, $\sum_j \Psi_j[\omega_m] \equiv 1$ (measured deviation $< 1.2 \times 10^{-7}$; construction details in Supplementary Section S-4). Each shell feeds a causal 1D convolution ($k \in \{15, 9, 5, 3\}$ coarse-to-fine), recombined by learned scale routing. Because the shells overlap smoothly at band edges, the front-end is *less* frequency-localized than standard wavelet banks: per-band in-octave spectral mass is $\{0.50, 0.33, 0.20, \approx 1.00\}$ versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) (filterbank frequency diagnostics). We therefore describe it as a Fourier-domain multiscale front-end, not an exact Littlewood–Paley octave decomposition, and measure its empirical contribution by matched ablation (Table Ib).

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [6]: excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \qquad (5)$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we evaluate an empirical heuristic multiplier $\tau_{\text{eff}} = \tau \cdot [1 + \frac{1}{6}\tanh(\gamma_2)]$ governed by the sample excess kurtosis γ_2 of calibration residuals: on heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} rises to suppress volatile noise; on light-tailed ones, it tightens for sensitivity.

We emphasize that this is an engineering heuristic for pre-asymptotic finite-sample tails, not an asymptotic EVT theorem. The primary deployed pipeline relies on standard SPOT ($u = p_{98}, q = 10^{-3}$) without the multiplier; the kurtosis heuristic is evaluated strictly as an explicit sensitivity arm (Section VII-D, Table III), where it degrades out-of-sample risk control (5.26% → 17.68% FPR) due to threshold lowering on non-positive kurtosis blocks.

More broadly, static EVT does not eliminate threshold-calibration difficulty: under nominal design risk $q = 10^{-3}$ (idealized asymptotic FPR 0.1%), measured held-out nominal FPR settles at 5.3–6.2% for the JEPAs and ≈ 22% for TimesNet/TranAD on the 33-stream cell-pooled buffered cohort (7.6–8.6% vs. ≈ 30% on the 19-stream macro cohort, Table II). This severe overshoot stems from tail-sample scarcity (median 21 calibration exceedances) and non-stationary distribution drift between nominal calibration and deployment. Static peaks-over-threshold calibration cannot guarantee dependable

risk control on drifting telemetry. The primary operational finding is that OpEntropy JEPA achieves substantial false-alarm suppression under static calibration, even while non-stationary drift prevents static extreme-value theory from guaranteeing exact nominal risk control. Thresholds use static EVT on held-out nominal splits; dynamic rolling SPOT is evaluated separately in the operational incident analysis.

V. Experimental Protocol and Evaluation Metrics

The master experiment matrix enumerating all evaluation suites (stream counts, partitions, horizons, calibrators, metrics) is in Supplementary Table S1.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated TCN (3 residual blocks, channel depth F=48, kernel 5, dilations $\{1, 2, 4\}$); global temporal average pooling aggregates the full C=256 context, with LayerNorm and dropout (p=0.20). The momentum target encoder follows EMA tracking with cosine schedule m_{base}=0.996 $\rightarrow$ $m_{\max}$=0.9995.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\min} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%), *nominal calibration* (next 17.5%), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap). The training block optimizes representation weights alone. Withheld from training, the nominal calibration block fits per-stream GPD tail parameters σ, ξ above u=0.98 at q=10^{-3}, estimating γ_2 strictly within this interval. The nominal evaluation block then measures held-out FPR on telemetry unobserved during threshold selection. Thresholds remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Supplementary Section S-5).

b) Calibrator Assignment Across Experiments: To avoid ambiguity, we specify the calibrator per experiment. *Primary held-out benchmark* (Tables II, III, Supplementary Table S31; all paired statistics): standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier); headline metrics are the "SPOT" column of Table III, with calibrator variants compared on identical score streams. *Ablations and controlled studies* (Section VI): standard SPOT ($u = p_{98}$, $q = 10^{-3}$), matching the primary benchmark; the kurtosis multiplier is evaluated strictly as an explicit sensitivity arm (Section VI-D, Table III). *Buffered-attribution and EVT mechanism studies*: standard SPOT unless the calibrator is the object of comparison. *Incident metrics* (Supplementary Section S-5): rolling SPOT. No experiment mixes calibrators across models within a table.

c) Multi-Seed Replication, Primary Endpoints, and Multiplicity Controls: Principal models on the 19 non-GHL streams are trained across five seeds (42, 123, 456, 7, 2024); the 14 large GHL streams ($> 10^6$ steps) use three seeds (five on the two smallest) for cost, and ablation controls use three. We report held-out nominal FPR, Point-F1, and PR-AUC as distinct outcomes; nonsignificant detection differences are not evidence of parity. The post hoc operating-curve analysis uses only streams with archived score dumps (19 buffered, 18 causal), not the full 33-stream benchmark.

Inference reports stream-macro means, seed SDs, and two-sided Wilcoxon signed-rank tests. We distinguish two resampling levels: within each stream, the held-out nominal evaluation split uses a moving-block bootstrap (L=64) over the chronological false-alarm indicator sequence to construct per-cell FPR confidence intervals under serial autocorrelation; across streams, paired model contrasts resample stream differences (2,000 stream-level bootstrap iterations with seeds averaged within stream). The independent statistical unit is the telemetry stream (n=19 non-GHL at five seeds; n=33 where GHL enters at available coverage), with dataset-clustered checks reported to account for shared sensor environments. For general detector contrasts, nonsignificant tests do not establish equivalence; instead, we report paired differences with effect sizes and CIs. Across general contrasts, p-values are Benjamini–Hochberg FDR-adjusted at $q_{\text{FDR}} = 0.05$; CIs spanning zero or adjusted $p > 0.05$ are treated as inconclusive. For physical regularizer variants, where hypotheses test whether additions alter baseline performance, we complement Wilcoxon tests with formal Two One-Sided Tests (TOST) against TS-JEPA: the null hypothesis $H_0 : |\mu_1 - \mu_2| \geq \Delta$ tests non-equivalence against $H_1 : |\mu_1 - \mu_2| < \Delta$ under an equivalence margin of $\Delta = \pm 0.05$ Point-F1. Supplementary Table S32 maps every reported test to its endpoint, cohort, paired-stream count, seed treatment, raw and adjusted p, and effect-size interval.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from implementation-specific discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline with identical windowing ($C = 256, S = 64$), standardization, and EVT calibration. TimesNet and TranAD are in-house re-implementations configured to published architectural depths and evaluated under a shared optimization budget (AdamW, $\eta = 10^{-3}$, cosine decay, 20 epochs); no method received per-model hyperparameter tuning beyond the shared configuration, ensuring that differences reflect representation and scoring dynamics rather than disparate tuning budgets. While this standardized budget ensures equal training exposure, it represents a controlled benchmarking standard rather than bespoke per-model search. As documented in Supplementary Section S-8 (Table S28), these

baseline implementations replicate published Point-Adjusted F1 scores on benchmark datasets (e.g., SMD and MSL), and native shorter-horizon evaluations ($W \in \{10, 96\}$) confirm that long windows did not handicap them. Hardware-specific throughput, batch, warm-up, and per-timestep latency measurements are reported in Supplementary Table S12.

Baseline Architectures and Parameter Capacities: We compare four architectural classes under controlled parameter scaling: (1) *Operator-Entropy JEPA* (proposed architecture, 285,733 parameters at K=25, -1.15% vs. unconstrained); (2) *TS-JEPA* (unconstrained latent control, 289,061 at K=25; 218,309 at K=9); (3) *physical controls*: Reynolds-Stress (295,953, $+2.4\%$) and Potential-Flow (338,117, $+17.0\%$); and (4) *observation-space reconstructive baselines*: TimesNet [5] (648,553 at K=25; 574,529 at K=9) and TranAD [14] (205,746 at K=25; 201,090 at K=9). To isolate whether deep representation is necessary, we evaluate a ridge-regularized *Linear AR Forecaster*. Non-parametric and score-bound controls (observation-space TCN, PCA, max-z, IsolationForest, kNN, MAD, and PCA→32→LayerNorm) are detailed in Supplementary Table S23.

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [28], [36] document severe reporting vulnerabilities in this literature (point-adjustment inflation, channel-averaging imbalances, threshold gaming), so we report metrics across complementary tiers. *Exploratory operating characteristics*: event recall integrated over nominal calibration budgets and, separately, over a fixed exploratory realized-test-FPR interval $[0.10, 0.50]$ (Section VII). For the latter, we interpolate within each stream–seed curve only when both bounds are observed, pair models within identical stream–seed cells, average paired differences by stream, and use dataset-level sign permutations and dataset-clustered bootstrap intervals; Benjamini–Hochberg adjustment covers all tested model, mapping, and endpoint combinations. Missing support is reported, not extrapolated. *Secondary endpoints*: Dataset-Macro Point-F1 (unweighted mean across datasets, preventing the 14-stream GHL block from dominating); channel-macro Point-F1; PR-AUC (average precision, ranking quality independent of calibration); event recall (fraction of contiguous anomaly segments containing ≥ 1 alarm); mean detection delay Δt (onset-to-first-alarm lag over detected events only); and the trivial all-positive reference $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at base rate π; models below it underperform a constant alarm. Pooled micro-F1 exposes flood-prone models; contiguous PA-F1 and test-sweep PA-F1 quantify the residual calibration gap.

D. Evaluation Regimes: Zero-Lookahead Causal vs. Buffered Inference

To keep the buffered-attribution regime separate from future-observation access, we evaluate two regimes: *strictly causal online detection* (at decision time T only $\{X_\tau\}_{\tau \leq T}$ is accessible; scores use trailing attribution; zero lookahead) and *buffered window detection* (S=64 future timesteps are buffered before scoring, capturing full trajectory evolution at a deliberate 64-step latency). Both record Mean Detection Delay, Event Recall within delay budgets ($\Delta t \leq 10, 64$), and false alarms per 1,000 steps ($\text{FAR}_{1\text{k}}$).

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone. Because controls run three seeds against the five-seed principal runs, the paired control means below are computed on the shared three-seed subset and differ slightly from the five-seed Table I(a) rows:

1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$:** Disabling the spectral regularizer leaves Point-F1 unchanged (0.241 ± 0.014 vs. 0.244 ± 0.024; difference -0.003, 95% interval $[-0.011, +0.005]$, Wilcoxon $p = 0.62$); PR-AUC stays 0.347–0.348.
2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$:** Disabling the stress loss yields 0.203 ± 0.059 vs. 0.208 ± 0.071 ($+0.005$, $[-0.006, +0.016]$, $p = 0.41$). Covariance estimator comparisons across nine streams (Supplementary Table S8) demonstrate estimator invariance: instantaneous, temporal-window, and batch-empirical covariances shift Point-F1 by $|\Delta\text{F1}| \leq 0.010$ (non-significant).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing shifts Point-F1 to 0.129 from 0.096 on the shared three-seed subset ($p_{\text{raw}} = 0.048, p_{\text{BH}} = 0.24$; 0.089 at five seeds in Table I(a)); both underperform TS-JEPA (0.234, $p_{\text{raw}} = 0.013, p_{\text{BH}} = 0.24$, non-significant under FDR adjustment). Noise-matched queries and ODE integration tests (Supplementary Table S27) confirm that performance deficits are not an artifact of train/test query mismatch. Numerical field diagnostics confirm that while the unprojected spatial gradient is curl-free (antisymmetric fraction $< 10^{-7}$), spherical tangent projection introduces a 15.0%–15.7% antisymmetric curl component (0.150 on SMAP `P-3`, 0.157 on MSL `M-1`).

Across the full 19-stream suite, the physical regularizers produce no statistically detectable improvement over unconstrained latent prediction. Formal Two One-Sided Tests (TOST) reject non-equivalence ($H_0 : |\Delta| \geq 0.05$) and confirm statistical equivalence to TS-JEPA within a tight $\Delta = \pm 0.05$ Point-F1 margin: Operator-Entropy achieves $p = 0.0009$ (95% CI $[-0.014, +0.021]$, also passing $\Delta = \pm 0.03$ with $p = 0.0283$) and Reynolds-Stress achieves $p = 0.0054$ (95% CI $[-0.042, +0.007]$). These null results establish practical equivalence rather than an empirical advantage.

B. Ablation 2: Auxiliary Input-Channel Gating Variant

The auxiliary causal input-channel gate is compared against uniform pooling and non-causal controls under an identical TS-JEPA configuration (Supplementary Section S-3, Tables S3, S4). The effect is stream-dependent and macro Point-F1 favors uniform weighting (0.118 vs. 0.077): gating roughly doubles

recall on transient faults at $\sim 3\times$ the nominal FPR, while quiet faults remain weakly detected under every regime (recall 0.02–0.07). It is therefore an evaluated variant; the deployed pipeline retains uniform input weighting.

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Replacing the standard front-end with the Fourier-domain dyadic shells plus per-shell causal convolutions and learned routing yields a modest, stream-dependent effect (Table I(b)): macro buffered Point-F1 improves from 0.088 to 0.131 across a five-stream exploratory panel (including two diagnostic streams, SMD and GECCO, outside the 33 primary streams), concentrated on MSL and SMD, with a small SMAP regression. Two caveats: the block adds $\approx 1.6 \times 10^5$ parameters, and measured shell overlap means octave isolation is not attained. The front-end is a modest contributor, not the primary driver.

D. Ablation 4: Threshold Calibration Rules on Nominal Splits

The full calibrator comparison is in Section VII-D; two findings bear on the ablations: non-parametric rules match SPOT (validation-max FPR 0.0553 vs. 0.0526 at equal F1), and the kurtosis-adaptive multiplier degrades calibration ($0.0526 \rightarrow 0.1768$ FPR at indistinguishable F1).

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

To discriminate the operational mechanisms driving tail conditioning and threshold transfer, all 19 non-GHL streams were re-run under the primary held-out protocol (five seeds, identical windows, AdamW budget, and SPOT $u{=}0.98/q{=}10^{-3}$ calibration) across controlled backbone arms sharing the `HybridTCNEncoder` ($C{=}256, S{=}64$, $d{=}32$, three dilated TCN layers): TCN-Obs-Recon (observation reconstruction), TCN-Obs-Pred (observation forecasting), JEPA-CtxRecon (latent autoencoding of EMA-encoded *context* without foresight), JEPA-NoEMA (latent prediction with hard-copy target), and TS-JEPA. Five further controls probe score geometry and gating directly: TCN-ObsPred-LN (same forecaster, LayerNorm on the 32-dimensional bottleneck and flattened forecast, identical saliency gate), PCA-LinPred-LN, Rand-LinPred (fixed random Gaussian projection to 32 dimensions), Obs-MAD, IsoForest, KNN-Dist (training-free scorers), and TS-JEPA-NoGate (identical model, ungated ℓ_2 discrepancy). Supplementary Table S19 reports per-arm residual diagnostics; the earlier 15-stream multi-SNR study is in Supplementary Table S2.

Mechanism Discrimination: Foresight vs. Geometry vs. Dynamics: The controls are consistent with three distinct operational factors: (1) *Future temporal forecasting is not uniquely necessary for score conditioning*: JEPA-CtxRecon autoencodes context representations without forecasting future timesteps, yet achieves an FPR reduction of 0.202 ($0.310 \rightarrow 0.108$ relative to TCN-Obs-Pred) and higher event recall (0.513 vs. 0.473 for TS-JEPA on this controlled panel, Table S19; 0.481 on primary benchmark Table II). Latent representation compression, rather than future-state extrapolation, conditions scores for tail estimation. (2) *Bounded score geometry mechanically suppresses tail kurtosis (γ_2)*: projecting through a low-dimensional LayerNorm bottleneck compresses residual excursions regardless of architecture, as TCN-ObsPred-LN suppresses kurtosis to $\gamma_2 = 0.14$ and PCA-LinPred-LN achieves 0.87 (vs. 8.24 for unnormalized TCN-Obs-Pred and 1.14 for TS-JEPA). (3) *Representation dynamics and conservative subspace transfer*: kurtosis suppression alone cannot prevent threshold transfer breakdown across temporal blocks. TCN-ObsPred-LN suffers held-out nominal FPR of 0.186 (buffered) and 0.117 (causal trailing), versus 0.080 and 0.014 for TS-JEPA. At conservative budgets, PCA-LinPred-LN and IsoForest collapse to event recalls of 0.665 and 0.631 versus 0.774 for TS-JEPA (0.768 ungated). Representation rank diagnostics across diverse streams reveal that z_{ctx} and z_{tgt} occupy an effective subspace of 5.6–6.1 and 8.5–14.0 dimensions out of 32 (with feature variances 0.003–0.031 and leading PC share $\approx 53\%$). While ruling out rank-1 collapse, this compact subspace structure together with JEPA-CtxRecon's matched geometry (5.62 ctx rank, 0.513 recall) indicates that low nominal FPR is consistent with conservative subspace projection and bounded score geometry rather than an exclusive property of predictive temporal dynamics.

Target EMA and Window Horizons: Removing the EMA target changes little in Point-F1 (JEPA-NoEMA: -0.033 vs. TS-JEPA, $p{=}0.17$; spectral diagnostics show higher top-PC concentration); PR-AUC differences across arms are inconclusive ($p \geq 0.16$). Native-window checks (Supplementary Table S18) confirm that baseline tail heaviness does not arise from the $W{=}320$ window: at native horizons, TranAD ($W{=}10$) spikes to $\gamma_2 = 448.25$ and TimesNet ($W{=}96$) reaches $\gamma_2 = 98.14$ (44.09% FPR), while TS-JEPA retains $\gamma_2 = 3.14$.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table II reports the comparison across 33 continuous telemetry streams under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$). Primary findings:

False Alarm Suppression on Held-Out Telemetry: Latent predictive scoring via Operator-Entropy JEPA and TS-JEPA substantially suppresses false alarms under static calibration compared with leading reconstructive baselines: on 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.076–0.086 versus 0.300 for TimesNet and 0.307 for TranAD (a four-fold false-alarm reduction). Across all 33 streams, the paired reduction is statistically significant: OpEntropy JEPA achieves -0.144 vs. TimesNet (CI $[-0.241, -0.065]$, $p = 0.006$) and -0.142 vs. TranAD ($p = 0.006$); TS-JEPA yields -0.153 ($p = 0.002$) and -0.151 ($p = 0.003$). Clustering by dataset ($N = 7$) yields $p = 0.016$–0.031, verifying that significance does not depend on within-dataset stream replication. The underlying mechanism is verified under the controlled backbone panel (Section VI-E).

Point-F1 and Continuous Ranking (PR-AUC) Uncertainty: False-alarm suppression does not establish superior anomaly detection. On the non-GHL subset TS-JEPA attains 0.234 Point-F1 / 0.323 PR-AUC versus 0.197/0.329 (TimesNet) and 0.130/0.287 (TranAD). Across all 33 streams the paired differences remain inconclusive: vs. TimesNet $+0.012$ F1 (CI

TABLE I

ABLATION STUDIES. (A) MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED BACKBONES, 19 NON-GHL STREAMS (POINT-F1 = STREAM-MACRO MEAN ± SEED SD; HELD-OUT FPR ON UNSEEN NOMINAL DATA). (B) DYADIC-SHELL FRONT-END ON/OFF UNDER IDENTICAL TS-JEPA CONFIGURATION ACROSS FIVE MULTIVARIATE EXPLORATORY STREAMS (INCLUDING DIAGNOSTIC STREAMS SMD AND GECCO). AUXILIARY INPUT-CHANNEL GATING AND FAULT-TYPE RECALL PANELS ARE IN SUPPLEMENTARY SECTION S-3.

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.234 ± 0.030	0.323	0.076	0.481
Operator-Entropy JEPA	0.237 ± 0.013	0.323	0.086	0.506
$\lambda_{\text{ent}} = 0$ control	0.244 ± 0.024	0.347	0.083	0.518
Reynolds-Stress JEPA	0.207 ± 0.052	0.307	0.078	0.434
$\alpha_{\text{stress}} = 0$ control	0.203 ± 0.059	0.311	0.066	0.414
Potential-Flow JEPA	0.089 ± 0.036	0.201	0.091	0.303
no-regime control ($K_r = 1$)	0.129 ± 0.061	0.265	0.081	0.392
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream	Standard Conv	Dyadic Shells	Δ
SMAP `P-3`	**0.294**	0.261	−0.033
MSL `M-1`	0.002	**0.052**	+0.050
SMD `machine-1-2`	0.000	**0.177**	+0.177
Daphnet `S01R01E1`	0.144	**0.165**	+0.021
GECCO `water_quality`	0.000	0.001	+0.001
Macro Mean	0.088	**0.131**	+0.043

TABLE II

HELD-OUT CALIBRATION BENCHMARK, 33 TELEMETRY STREAMS (7 DOMAINS). THRESHOLDS: SPOT (q=10^{-3}) ON HELD-OUT NOMINAL SPLITS. METRICS: HELD-OUT NOMINAL FPR (ON DATA THE THRESHOLD NEVER SAW), UNADJUSTED POINT-F1, PR-AUC, EVENT RECALL, MEAN DETECTION DELAY Δt (AVERAGED STRICTLY OVER DETECTED ANOMALY EVENTS WITH EVENT RECALL > 0; SCORES EMITTED UPON TARGET WINDOW COMPLETION $\tau = t + S - 1$, $S = 64$). "NON-GHL": STREAM-MACRO OVER 19 STREAMS AT FIVE SEEDS; "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS; "DATASET-MACRO": UNWEIGHTED MEAN OF THE 7 DATASET MEANS. BEST IN **BOLD**, SECOND UNDERLINED.

	Non-GHL Streams (19 Streams, 5 Seeds)					Summary	Gasoil Heating Loop (14 Streams)	
Model Architecture	**Held-Out FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Mean Delay (Δt) ↓**	**DS-Macro F1 ↑**	**Held-Out FPR ↓**	**Point-F1 ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.076	0.234	0.323	0.481	80.3	**0.148**	0.005	0.000
Operator-Entropy JEPA	0.086	**0.237**	0.323	0.506	192.3	0.130	0.012	0.001
Reynolds-Stress JEPA	0.078	0.207	0.307	0.434	132.8	0.122	**0.003**	0.002
Potential-Flow JEPA	0.091	0.089	0.201	0.303	146.9	0.065	0.011	0.006
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.300	0.197	**0.329**	**0.583**	**48.5**	0.100	0.061	**0.022**
TranAD (VLDB 2022)	0.307	0.130	0.287	0.490	234.4	0.069	0.046	0.020
Non-Neural Linear Baseline								
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	0.038	**0.002**	0.000

$[-0.053, +0.075]$, p=0.54), -0.000 PR-AUC (p=0.39); vs. TranAD $+0.051$ F1 (p=0.88), $+0.023$ PR-AUC (p=0.16). Under matched operational calibration budgets (Supplementary Table S21), latent prediction trades event recall (0.486 at q^*=0.001) for precision (0.277 vs. 0.194 TimesNet) and lower alarm frequency (207.9 vs. 259.2 per 1,000 steps). Detection superiority is unproven on point-level metrics.

Event Recall and Detection Conservatism: Lower FPR generally accompanies lower event recall: on non-GHL streams TimesNet captures 58.3% of events, while OpEntropy JEPA attains 50.6%, exceeding TranAD (49.0%), TS-JEPA (48.1%), Reynolds (43.4%), and Potential-Flow (30.3%). Subtle onsets that do not shift dominant latent coordinates are filtered with the noise.

Detection Delay Trade-Offs (Δt): Mean detection delay (Table II) varies widely: TimesNet responds fastest (Δt=48 steps), TS-JEPA averages 80, and the physics variants are slower still (133–192); TranAD is slowest (234). This raw disparity is largely a detected-set artifact, as delays are averaged across different event sets per model. On shared intersecting detected events at q^*=0.05, the delay gap between TimesNet and TS-JEPA collapses from 31.8 steps to just 2.3 steps (56.9 vs. 59.2 steps; Table S24b); vs. TCN-ObsPred it is 60 vs. 51 steps. The apparent latency gap reflects which events each detector catches rather than the intrinsic response delay.

Necessity of Nonlinear Representation Learning (Linear AR Control): A ridge-regularized linear AR forecaster under identical calibration attains FPR 0.005 but collapses to Point-F1 0.051: observation-space linear dynamics cannot separate faults from background variance. Two training-free references sharpen the point (Supplementary Table S23): a PCA-subspace reconstructor matches TimesNet almost exactly (0.311/0.198/0.331 vs. 0.300/0.197/0.329 FPR/F1/PR-AUC), while a per-channel max-z scorer attains the best ranking (0.374) at the worst FPR (0.339): ranking quality and threshold conditioning are independent properties.

Domain Dynamics, Failure Modes, and Regimes of Applicability: Detection performance varies across physical environments (Supplementary Tables S16–S17), defining three operational regimes: (1) *Favorable regimes (latent conditioning succeeds)*: High-dimensional telemetry with continuous subsystem dynamics and structured correlations (NASA SMAP and MSL rovers) favors latent predictive scoring. On SMAP, Operator-Entropy attains 0.591 PR-AUC; on MSL, TS-JEPA attains 0.445 vs. 0.159 for TimesNet, decisively exceeding the trivial all-positive detector ($\text{F1}_{\text{all-pos}} = 0.243$ and 0.396; Table S24a). (2) *Challenging regimes (conservative missed events)*: Quiet faults (flatlines, sensor freezes, stuck actuators) or subtle low-frequency drifts that do not alter dominant latent coordinates are filtered alongside ambient noise. On these events, latent predictors exhibit lower event recall than unconstrained reconstructors. (3) *Adverse regimes (defeats all evaluated models)*: On high-frequency process noise or continuous non-stationary baseline ramps (GHL heating loop, Metro traffic volume), all deep detectors struggle (0.000–0.022 Point-F1). On high-prevalence streams (Daphnet kinematics π=14.8%; SWAN magnetograms π=28.0%), deep detectors perform near or below a constant alarm rule ($\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$), confirming that non-stationary regime shifts

violate static calibration assumptions across all architectures.

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

Does adding physics-inspired regularizers (Koopman spectral entropy, Reynolds-stress closure, curl-free potential flow) improve detection over unconstrained TS-JEPA? The 33-stream data do not support an aggregate gain:

1) **Operator-Entropy vs. TS-JEPA:** paired difference $+0.002$ Point-F1 (CI $[-0.025, +0.029]$, p=0.71), $+0.002$ PR-AUC (p=0.86), $+0.009$ FPR (p=0.75); leads on 9 of 33 streams.
2) **Reynolds-Stress vs. TS-JEPA:** -0.015 Point-F1 ($[-0.042, +0.007]$, p=0.61), -0.007 PR-AUC (p=0.13); leads on 11 of 33.
3) **Potential-Flow vs. TS-JEPA:** Point-F1 differs by -0.081 under SPOT (p=0.033 unadjusted). Its higher event-recall area over nominal causal budgets is descriptive; without matched realized-FPR support and multiplicity-adjusted evidence, neither an intrinsically stronger detector nor a purely operating-point explanation is established.

These findings align with the matched-capacity controls in Table I(a): the physical regularizers provide no aggregate detection advantage over unconstrained latent prediction. The dynamical diagnostics on *why* (operator rank preservation without detection gain, covariance-shift insensitivity, and decoupling resolution via Lyapunov invariance) are in Supplementary Section S-4.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Event-level incident metrics, latency, and rolling-SPOT case studies are in Supplementary Section S-5: rolling SPOT suppresses false alarms up to $21\times$ on drift-prone streams at a recall cost, and TS-JEPA's causal false-positive rate on the 5-stream operational incident cohort (evaluated across test-split nominal intervals: 4.18%, Table S8) sits an order of magnitude below reconstructive baselines (20.8–30.1%). In comparison, on the primary 19-stream held-out calibration benchmark, held-out nominal FPR is 0.0136 vs. 0.2427/0.2219.

Supplementary Table S31 reports the same benchmark under strictly causal (0-lookahead) streaming. On the 19 non-GHL streams, TS-JEPA reduces held-out nominal FPR to 0.0136 versus 0.2427 (TimesNet) and 0.2219 (TranAD), an $\approx 18\times$ suppression; on the 14 GHL streams, causal FPR falls below 0.009 for JEPAs, but causal Point-F1 there is near-zero for latent predictors (≤ 0.006 vs. 0.022 TimesNet). Causal Point-F1 drops to 0.1118 ± 0.0412 (TS-JEPA) versus 0.2091 ± 0.0265 (TimesNet), and causal event recall is 46.55% for TS-JEPA versus 58.82% for TimesNet: without a lookahead buffer, latent predictive scoring is an ultra-conservative filter that prevents false alarms at the price of detection sensitivity.

a) Pre-Onset Alarm Attribution: Buffered attribution is strictly retrospective: scores are painted over $[t{+}1, t{+}S]$, precluding leads beyond S=64 steps. Median pre-onset "leads" of 180–204 steps in empirical lead-time diagnostics are nuisance alarms at the underlying false-positive rate or precursor drift,

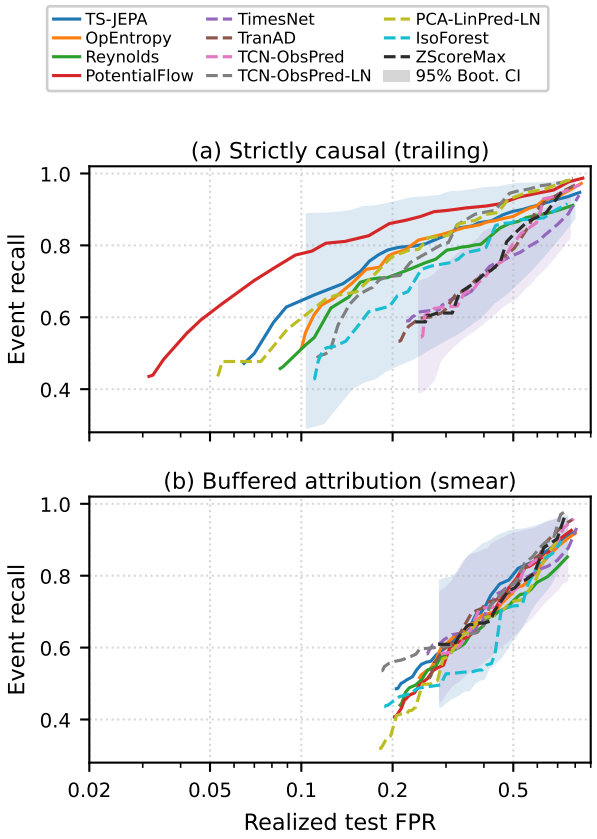


Fig. 3. Pooled event recall versus realized test FPR at common nominal calibration budgets: (a) causal trailing; (b) buffered smear. Bands resample datasets; pooled curves are descriptive, not paired estimates at matched FPR.

not prediction: TimesNet emits 206 such alarms per 1,000 nominal steps versus 24 for TS-JEPA, reflecting the baseline alarm-rate gap. We do not claim early-warning capability.

b) Detection Quality at Matched Operating Points: We fit empirical-quantile thresholds on nominal budgets $q \in [10^{-4}, 0.9]$ to evaluate event recall and realized FPR on test. Partial-AUC values 0.742 (TS-JEPA) and 0.618 (TimesNet) integrate recall over nominal budgets up to 0.10, not matched realized FPR. At q^*=0.05, causal event recall is 0.774 vs. 0.618, but realized FPR differs (0.180 vs. 0.262). Budget-pooled curves (Fig. 3) obscure stream-specific support and are not paired detector comparisons. Given multi-hundred-step anomalies (mean 402 in Daphnet, 1,102 in SMAP), alarming at 30% FPR catches $> 95\%$ of segments by chance $(1 - (1 - \text{FPR})^L)$; TimesNet's 58% recall at 30% FPR falls within this random expectation.

The central operational question (whether latent prediction detects more events than competing baselines at comparable realized alarm rates) remains unresolved on the available telemetry streams due to limited common-support coverage and statistical power. For an exploratory support-aware check, we interpolate stream–seed curves over common realized-FPR $[0.10, 0.50]$ without extrapolation (Supplementary Table S30). The causal TS-JEPA–TimesNet difference is $+0.061$ (95% dataset-bootstrap CI $[-0.153, 0.329]$; dataset-sign-permutation p=0.625; 20 paired cells across only four streams). For TS-JEPA vs. PCA-LinPred-LN ($+0.121$), the pre-specified con-

TABLE III
CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS (33 STREAMS, CELL-WEIGHTED MEANS WITH STREAM-MACRO IN PARENTHESES; UP TO FIVE SEEDS). DESIGN RISK q=10^{-3}=0.1%. OVERSHOOT $> q$ IS THE SHARE OF CELLS WHOSE 95% CI LOWER BOUND EXCEEDS $q = 10^{-3}$. BUFFERED SMEAR INFLATES FPR BY PAINTING ALARMS OVER $S = 64$ STEPS; CAUSAL TRAILING REFLECTS STRICT STEP-LEVEL ALARMS.

Model Architecture	Calibrator Rule	Causal FPR	Overshoot $> q$	Buffered FPR	Point-F1 (buf)
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	0.0101 (0.0092)	7%	0.0526 (0.0457)	0.1577
	SPOT + Kurtosis ($\beta = 0.167$)	0.0108 (0.0094)	7%	0.1768 (0.1515)	0.1694
	Empirical 99.5th Percentile	0.0169 (0.0157)	21%	0.0682 (0.0595)	0.1598
	Validation Maximum	0.0090 (0.0080)	6%	0.0553 (0.0478)	0.1591
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	0.0152 (0.0144)	11%	0.0617 (0.0546)	0.1599
	SPOT + Kurtosis ($\beta = 0.167$)	0.0136 (0.0125)	9%	0.1229 (0.1064)	0.1441
	Empirical 99.5th Percentile	0.0238 (0.0229)	25%	0.0758 (0.0677)	0.1658
	Validation Maximum	0.0148 (0.0136)	7%	0.0656 (0.0576)	0.1627
TimesNet	Standard SPOT ($q = 10^{-3}$)	0.1795 (0.1611)	49%	0.2217 (0.1985)	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.1843 (0.1623)	46%	0.2281 (0.2011)	0.1352
	Empirical 99.5th Percentile	0.1914 (0.1752)	51%	0.2338 (0.2125)	0.1413
	Validation Maximum	0.1791 (0.1599)	48%	0.2263 (0.2032)	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	0.1601 (0.1423)	46%	0.2217 (0.1963)	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.1722 (0.1513)	43%	0.2453 (0.2151)	0.0940
	Empirical 99.5th Percentile	0.1712 (0.1544)	51%	0.2343 (0.2093)	0.0962
	Validation Maximum	0.1580 (0.1392)	44%	0.2270 (0.2009)	0.0952

firmatory test is the stream-level paired Wilcoxon test and 95% bootstrap CI ($p = 0.312$, $[-0.008, 0.271]$, 11 streams); 5-dataset tests are low-powered exploratory checks ($p \geq 0.0625$; permutation $p = 0.188$, $p_{\text{BH}} = 1.0$; cluster bootstrap CI $[0.015, 0.310]$ reflects $N = 5$ undercoverage). TS-JEPA establishes no significant advantage over PCA-LinPred-LN on common support. Bounded-observation TCN matches TS-JEPA causally on nine streams (-0.018, CI $[-0.055, 0.003]$) and shows higher buffered recall area on five streams (-0.211, CI $[-0.349, -0.105]$); neither survives family-wide BH adjustment. Nominal-budget orderings and pooled curves do not establish a causal-regime advantage, and boundedness alone determines neither tail transfer nor matched-FPR detection.

D. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does automated extreme-value thresholding adhere to its design risk q out of sample? Table III compares calibrators on identical score streams across all 33 streams under both strictly causal trailing and buffered smear inference. The calibrator evaluated here is the *static* peaks-over-threshold step of SPOT: a single threshold fit once on the calibration block; dynamic rolling SPOT is evaluated in Supplementary Section S-5. Four realities emerge:

1) **Causal vs. buffered smear overshoot:** Under causal trailing, TS-JEPA achieves held-out nominal FPR of 0.0101 (0.0092 stream-macro; median 0.0002, IQR $[0.0000, 0.0025]$). Buffered inference smears each alarm across S=64 steps, inflating per-timestep FPR by up to $S \cdot q = 6.4\%$ to reach 0.0526 (0.0457 stream-macro). In contrast, TimesNet and TranAD suffer causal FPRs of 0.1795 and 0.1601 (0.2217 buffered), exceeding the smear ceiling and confirming distributional breakdown.
2) **One-sided overshoot testing:** Rather than two-sided CI exclusion (counting zero-FPR cells as failures), one-sided testing ($\text{CI}_{\text{lo}} > 10^{-3}$) reveals TS-JEPA overshoots design risk in only 7% of causal cells (12% buffered), versus 46–49% for reconstructive baselines.
3) **Parity with empirical and conformal quantiles:** On TS-JEPA, validation-maximum yields causal FPR 0.0090 (F1 0.0722), empirical p99.5 yields 0.0169 (F1 0.0907), and Split Conformal thresholding (α=10^{-3}) achieves median causal FPR 0.0006 vs. 0.1247 (TimesNet) and 0.1866 (TranAD). Neither parametric GPD nor conformal calibration prevents risk overshoot because non-stationary drift violates exchangeability.
4) **Kurtosis-adaptive directional breakdown:** The heuristic multiplier $1 + \frac{1}{6}\tanh(\gamma_2)$ inflates TS-JEPA buffered FPR from 0.0526 to 0.1768: for $\gamma_2 \leq 0$ (33.3% of cells), it lowers thresholds and raises FPR from 9.17% to 50.49%; where $\gamma_2 > 0$ (66.7%), it raises thresholds and reduces FPR from 3.31% to 1.28%.

Empirical tail diagnostics on nine non-GHL streams confirm tail-sample scarcity: median exceedances above $u = p_{98}$ is only 21 (range 16–70), and Grimshaw returns degenerate $\xi = -0.9$ in 33% of TS-JEPA and 52% of TimesNet fits. Yet KS tests reject in only $6/27$ (TS-JEPA) and $8/27$ (TimesNet) cells, while held-out FPR varies from 0.0 to 0.67 across seeds on the same stream under passing fits.

To determine whether drift or scarcity dominates, an exchangeable permutation control uniformly samples calibration and evaluation splits from pooled nominal blocks. Under point exchangeability, causal TS-JEPA FPR drops from 0.0121 ($12.1\times$ overshoot) to 0.0019 ($1.9\times$ overshoot), matching design q=10^{-3} within sampling variance; TimesNet drops from 0.0872 to 0.0037 (a 96% overshoot reduction). Block permutation (L=64) yields intermediate overshoot ($5.8\times$ for TS-JEPA), indicating that inter-block drift dominates calibration failure under the tested permutation protocol while autocorrelation is secondary. Across all $N = 1080$ dumped cells, KS drift between calibration and test nominal blocks predicts realized SPOT FPR (Spearman $\rho = +0.377$, 95% CI $[+0.314, +0.433]$, $p = 7.4 \times 10^{-38}$); calibration-block excess kurtosis correlates inversely ($\rho = -0.315$, 95% CI $[-0.373, -0.260]$, $p = 2.3 \times 10^{-26}$). A controlled-drift study (Supplementary Table S22) shows runs declustering leaves only 5–9 effective exceedances per cell, and rolling calibrators control risk to 1–7% under step, variance, and autocorrelation shifts but fail under continuous ramps (0.33–0.39 FPR). Profiling shows JEPAs use 183.8–185.3 MB peak memory at 0.60–0.84 ms per window (Supplementary Table S12).

VIII. CONCLUSION AND FUTURE WORK

We formulated Operator-Entropy JEPA for multi-sensor anomaly detection and evaluated latent score conditioning, discriminative detection, and operational calibration across 33 telemetry streams under a strict held-out protocol:

1) **Operator-Entropy JEPA and False-Alarm Suppression:** OpEntropy JEPA suppresses nominal false alarms under static calibration by four- to five-fold compared with reconstructive baselines (TimesNet, TranAD), resolving severe alarm flooding on complex telemetry (NASA SMAP/MSL). However, this conservatism does not establish universal detection dominance: paired Point-F1 and PR-AUC differences remain statistically comparable, and causal streaming without lookahead incurs an event recall penalty on quiet flatlines and subtle drifts.
2) **Empirical Evaluation of Physical Regularizers:** The tested regularizers provide no aggregate detection benefit over unconstrained TS-JEPA: Operator-Entropy and

Reynolds-Stress confirm formal TOST equivalence within ± 0.05 Point-F1 on the multi-seed suite ($p \leq 0.0054$; 95% CIs $[-0.014, +0.021]$ and $[-0.042, +0.007]$), establishing practical equivalence.

3) **Calibration Dynamics under Distribution Drift:** Static extreme-value calibration systematically overshoots nominal design risk under chronological deployment (5.3–6.2% vs. $\approx 22\%$ buffered; 1.0% vs. 16–18% causal at design $q{=}10^{-3}$). Under an exchangeable permutation protocol, inter-block distribution drift accounts for 84–96% of this calibration overshoot, proving that non-stationary partition shifts dominate sample scarcity and autocorrelation, highlighting the need for adaptive threshold tracking in operational deployments.

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